



BY AMETECs

MR-02

Consolidated Test Suite — Sessions of 11–12 August 2026

SmartEPT Team Test Cases · 120 cases · 63 P1 · 3 parts · companion to MR-01

Version 1.0

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How to Use This Suite

Part 1 (sessions 1–3 features) runs on **live after deploy**. **Parts 2–3** (per-tenant licensing; licence change flows) run on **local Laragon first**, then their P1s again on live. Priorities: **P1** must pass before sign-off, **P2** within the week, **P3** nice-to-verify. Record results in the Excel workbook (MR-02 .xlsx). **Sign-off rule: every P1 = Pass ⇒ the full 11–12 Aug body of work is approved on live. Any P1 Fail ⇒ STOP and report to Ejaz immediately.**

1A. Login, SMTP & forgot-password (S1, Product 22ed12e)

ID	P	Test	Steps	Expected
TC-1A01	P1	Seeded/unlocked super login works (Pre: Live deployed; super password unlocked via tinker and CHANGED)	1. Sign in at admin.smartept.com/admin as super@smartept.io with the new password	Login succeeds; Super Admin console loads (before the fix NO seeded login ever worked)
TC-1A02	P1	All seeded passwords changed (Pre: Live)	1. Attempt login for each seeded demo account with the old default 'password'	Every attempt FAILS — defaults were rotated after the double-hash fix
TC-1A03	P1	Global SMTP relay set and delivering (Pre: Super login; real SMTP credentials)	1. Ops → Email/SMTP → global section → save + Test	Test reports success; a real email arrives
TC-1A04	P1	Forgot-password end-to-end on live (Pre: Global SMTP set)	1. Console login card → Forgot password? → enter email → receive code → set new password 2. Sign in with it	OTP arrives by EMAIL (not log); reset works; login OK; audit shows PASSWORD_RESET_REQUESTED/COMPLETED
TC-1A05	P2	Company SMTP overrides global (Pre: A company with its own SMTP saved)	1. Save company SMTP (Ops card, company section) + Test 2. Trigger any mail for that company (e.g. forgot-password for its user)	Mail goes out via the COMPANY's SMTP/from; clearing host falls back to global
TC-1A06	P2	OTP guardrails	1. Request OTP for an unknown email 2. Enter a wrong OTP 5 times 3. Use an expired (>10 min) code	Neutral response for unknown email (no probing); 5 attempts then blocked; expiry enforced

ID	P	Test	Steps	Expected
TC-1A07	P2	Resend cooldown (S3, 389b8a0)	1. Request OTP → button becomes 'Resend code (60s)' counting down 2. Try resending immediately via API	Button disabled for 60s; server also refuses early resends (throttle)

1B. Live bug verifications (S2, Product 1f6be35)

ID	P	Test	Steps	Expected
TC-1B01	P1	Attendance derivation clean (Pre: Live deployed for 1+ day)	1. Check laravel.log for 'lessThan()' on string' 2. Open Attendance for previously-affected employees	Zero new occurrences; attendance rows derive correctly (self-healed on next run)
TC-1B02	P1	Rules screen saves CLOSE and BLOCK on both types (Pre: Admin login)	1. App & Web Rules → set a website rule action = CLOSE → Save 2. Set an application rule action = BLOCK → Save	Both save without error (previously 1265 'Data truncated' — ENUM strike four)
TC-1B03	P2	Archive restore works (Pre: An archived (deleted) employee exists)	1. Employees → Archives → Restore	Restore completes; no 'Undefined array key new_employee_code'

1C. SEO & landing (S2, Central 4 commits)

ID	P	Test	Steps	Expected
TC-1C01	P1	Sitemap is https-only allow-list (Pre: Live Central deployed)	1. Open https://smartept.com/sitemap.xml	Exactly 7 URLs, all https://smartept.com/... (/ , contact, privacy, terms, refunds, security, system-requirements); no /buy, /pay, /client, /admin
TC-1C02	P2	Canonicals + noindex headers	1. View source of / and /privacy → canonical tag 2. curl -I https://smartept.com/client/login and /buy	Self-referencing https canonicals; workflow pages return X-Robots-Tag: noindex, follow
TC-1C03	P1	GA4 flows on publish (Pre: GA4 id saved in Landing Editor)	1. Landing Editor → Publish → make live 2. View source of / , /privacy, /buy	Exactly ONE gtag install with the saved ID on each page; publishing twice never duplicates
TC-1C04	P2	www redirects in one hop (Pre: Manual Nginx step applied on VPS)	1. curl -I for http://www., https://www., http://smartept.com	Each answers a SINGLE 301 to https://smartept.com/ (no chains, no 502)

1D. Branded URLs, licence auto-config & agent fix (S3)

ID	P	Test	Steps	Expected
TC-1D01	P1	Agent works on a branded URL (Pre: Live product deployed)	1. Configure an agent with <code>https://admin.smartept.com/khan</code> 2. Sign in and sync	Login + sync succeed — <code>/khan/api/308-bounces to /api/</code> with method+body preserved
TC-1D02	P1	Licence validates with NO .env URL (Pre: Product server without <code>SMARTEPT_LICENSE_URL</code>)	1. Licence screen → Validate now	Phones <code>https://smartept.com</code> (Setting pushed by Central, else default) — no 'Central URL not configured' error
TC-1D03	P1	Cross-company device re-registration (agent-500 fix) (Pre: A PC previously registered under company A)	1. Sign the agent in as an employee of company B on the same PC	Registration succeeds (no HTTP 500 / duplicate-key); device re-points to B; audit <code>DEVICE_COMPANY_REASSIGNED</code> ; old company-A token dead (401)
TC-1D04	P2	Payment History ledger (Central a055ed3) (Pre: Client with partial payments)	1. Central → client modal → Payment History	Every payment row with method/reference + 'Total received' matching the ledger

1E. Money spot-checks (S1 pricing — regression guard)

ID	P	Test	Steps	Expected
TC-1E01	P1	Progressive lifetime price on /buy (Pre: Live bands (e.g. 15 → ₹15,000, 30 → ₹25,000))	1. /buy → On-Premise → 15, 22, 30 users	15=₹15,000; 30=₹25,000; 22 = interpolated between (~₹19,667 for the example bands); above last band = Custom Quote, never ₹0
TC-1E02	P2	Admin delete guards (73888fc) (Pre: Super login; one paid order, one clean quote)	1. Try Delete on the PAID order 2. Delete the clean quote	Paid/money-bearing order: NO delete offered; clean quote deletes with confirmation; audit entries written

1F. Test-system health (S5 — run by a developer)

ID	P	Test	Steps	Expected
TC-1F01	P1	Product full suite green (Pre: Repo at <code>da11247+</code>)	1. <code>cd C:\laragon\www\smartept && php artisan test</code>	0 failures (first healthy state since 19-Jul)
TC-1F02	P1	Central full suite green (Pre: Repo at <code>d28b56c+</code>)	1. <code>cd C:\laragon\www\smartept-central && php artisan test</code>	56 passed / 342 assertions, 0 failures
TC-1F03	P2	Provision endpoint tolerates omitted optional fields (2ca7dec) (Pre: Provisioning secret known (test env only))	1. <code>POST /api/provision</code> with only <code>external_tenant_id</code> , <code>company_name</code> , <code>admin_email</code>	201 — no 500 'Undefined array key timezone'

2A. Migration & data move — per-tenant licensing (S4)

ID	P	Test	Steps	Expected
TC-2A01	P1	Migration adds company_id and unique index cleanly (Pre: <i>Product repo at 1fcb7f7; DB backed up</i>)	1. php artisan migrate 2. DESCRIBE installation_licenses	Migration 2026_08_12_000400 runs without error; company_id column nullable BIGINT with UNIQUE index; no data lost
TC-2A02	P1	Khan key auto-moves to Khan's tenant row (Pre: <i>Install slot holds Khan's key; exactly one AMETECs_SAAS company named 'Khan Incorporation'</i>)	1. Run migration 2. SELECT company_id, license_key FROM installation_licenses	The existing row's company_id = Khan's company id; key, bundle, status preserved; no second row created
TC-2A03	P2	No-match safety (client-hosted shape) (Pre: <i>DB where bundle company name matches no AMETECs_SAAS company</i>)	1. Run migration on such a DB (e.g. local seeded)	Nothing moves: the licensed row keeps company_id NULL; behaviour identical to before
TC-2A04	P2	Ambiguity safety — two same-name SaaS companies (Pre: <i>Two AMETECs_SAAS companies with identical name; install slot keyed</i>)	1. Run migration	Move is skipped entirely (count != 1); no data changed
TC-2A05	P3	Migration down() reverses cleanly (Pre: <i>Migration applied</i>)	1. php artisan migrate:rollback --step=1	Unique index and company_id column dropped; table intact
TC-2A06	P1	Fresh install slot after key move (Pre: <i>Migration moved the key (Khan case)</i>)	1. Open Licence screen as Super Admin 2. Check evaluation end date	A fresh empty install row exists (created on first access); shows EVALUATION with 7 days from creation; agents of non-SaaS companies keep working during window

2B. Licence resolution model — per-tenant licensing (S4)

ID	P	Test	Steps	Expected
TC-2B01	P1	AMETECs_SAAS company governed by own row (Pre: <i>SaaS company exists</i>)	1. tinker: InstallationLicense::governing(\$saasCompany)	Returns row with company_id = that company; creates it empty on first call

ID	P	Test	Steps	Expected
TC-2B02	P1	Non-SaaS company governed by install row (<i>Pre: Company with deployment_model LAN/PRIVATE_CLOUD/HYBRID/null</i>)	1. tinker: InstallationLicense::governing(\$company)	Returns the install-level row (company_id NULL) — pre-existing behaviour
TC-2B03	P2	Null company (operator context) → install row	1. tinker: InstallationLicense::governing(null)	Returns install-level row
TC-2B04	P2	forCompany is idempotent (<i>Pre: SaaS company</i>)	1. Call InstallationLicense::forCompany(\$id) twice	Same row id both times; created_at unchanged (evaluation clock stable)
TC-2B05	P2	Unique index blocks duplicate tenant rows (<i>Pre: Tenant row exists</i>)	1. Attempt raw INSERT with same company_id	MySQL rejects with duplicate-key error; app never creates duplicates (firstOrCreate)
TC-2B06	P3	current() ignores tenant rows (<i>Pre: Tenant rows exist</i>)	1. tinker: InstallationLicense::current()	Returns only the company_id-NULL row, never a tenant's

2C. Licence screen by role — per-tenant licensing (S4)

ID	P	Test	Steps	Expected
TC-2C01	P1	Cloud tenant admin sees own licence with scope line (<i>Pre: Khan admin login (via /khan or /admin console)</i>)	1. Sign in as Khan COMPANY_ADMIN 2. Open Licence	Scope row: 'This licence belongs to Khan Incorporation — issued and managed by SmartEPT Central'; Khan's masked key, plan, seats, expiry shown
TC-2C02	P1	Tenant admin Save & validate writes own row only (<i>Pre: Khan admin; valid key from Central</i>)	1. Paste key, Save & validate 2. Check installation_licenses	Key stored on Khan's row (company_id set); install slot untouched; status ACTIVE after validation
TC-2C03	P2	Tenant admin Validate now phones Central for own key (<i>Pre: Khan row configured</i>)	1. Click Validate now 2. Check last_checked_at on both rows	Khan row refreshed; install row last_checked_at unchanged

ID	P	Test	Steps	Expected
TC-2C04	P2	Offline licence card hidden for cloud tenant admin (Pre: Khan admin on Licence screen)	1. Inspect page	'Offline licence file' card not visible; fingerprint/import controls absent
TC-2C05	P1	Offline .lic import API blocked for tenant scope (Pre: Khan admin token)	1. POST /api/license/import with any token	HTTP 403: 'Offline licence files apply to self-hosted servers...' message; no state change
TC-2C06	P2	Super Admin licence screen shows install slot + pointer (Pre: super@smartept.io)	1. Open Licence	Install-level licence (or EVALUATION) shown; note 'Cloud tenants on this install carry their own licences — see the Tenants screen'
TC-2C07	P1	Client-hosted admin behaviour unchanged (regression) (Pre: Local seeded install (LAN company)); admin@ameteecs.io)	1. Open Licence; save key; validate; import .lic	Identical to pre-change behaviour: install slot used, offline card visible and functional
TC-2C08	P2	Device seats counted per scope (Pre: Khan row active with devices; other companies have devices too)	1. Khan admin Licence screen: seats line 2. Super admin Licence screen: seats line	Khan sees only Khan devices registered; install scope counts all devices on the box

2D. Provisioning key delivery (Central → Product) — per-tenant licensing (S4)

ID	P	Test	Steps	Expected
TC-2D01	P1	New cloud tenant fully licensed hands-free (Pre: Central 76ddf09 deployed; product 1fcb7f7 deployed; provisioning secret set)	1. In Central create/pay a cloud tenant (or Save an active one) 2. Open product Tenants screen	Company + admin created AND licence row ACTIVE with the tenant's key — no manual key entry anywhere
TC-2D02	P1	Re-provision re-pushes the key idempotently (Pre: Existing provisioned tenant)	1. Save the tenant again in Central 2. Check product row	Same key → row untouched (status/bundle preserved); changed key → row reset and revalidated
TC-2D03	P2	Provision without licence_key still works (legacy) (Pre: Tenant with no active licence)	1. Provision from Central	Company + admin created; no licence row forced; tenant runs on 7-day evaluation
TC-2D04	P1	Bad provisioning secret rejected	1. POST /api/provision with wrong X-Provision-Secret	HTTP 401; nothing created or changed
TC-2D05	P2	Open my Console SSO unaffected (Pre: Provisioned tenant with console_url)	1. Central portal → Open my Console	Signed-in landing on branded console exactly as before

2E. Agent enforcement (EnsureLicensed) — per-tenant licensing (S4)

ID	P	Test	Steps	Expected
TC-2E01	P1	Licensed tenant agents sync normally (Pre: Khan licence ACTIVE)	1. Khan employee signs into agent; agent syncs	register-device / heartbeat / sync all pass
TC-2E02	P1	One tenant's dead licence never blocks another (Pre: Tenant X row set status=revoked (tinker); Khan ACTIVE)	1. X agent syncs 2. Khan agent syncs	X gets 403 LICENSE_BLOCKED (reason revoked); Khan completely unaffected
TC-2E03	P1	Expired within grace still syncs (Pre: Tenant row status=expired, expires_at within grace_days)	1. Agent sync	Traffic passes (grace window honoured per tenant)
TC-2E04	P2	Expired beyond grace blocks (Pre: Tenant row expired beyond grace)	1. Agent sync	403 LICENSE_BLOCKED for that tenant only
TC-2E05	P1	Unlicensed SaaS tenant: 7-day evaluation then block (Pre: SaaS company with no key (fresh row))	1. Sync during window 2. Set row created_at back 8 days (tinker); sync again	Passes during window; then 403 with reason evaluation_expired
TC-2E06	P2	Non-SaaS companies ride the install slot (Pre: Install slot ACTIVE (or in evaluation))	1. Agent of LAN company syncs	Governed by install slot exactly as before the change
TC-2E07	P2	Central unreachable → last verdict stands per tenant (Pre: Kill SMARTIPT_LICENSE_URL reachability (hosts file) on a test box)	1. Tenant with cached ACTIVE syncs	Sync continues; unreachable_since recorded; no false block

2F. Device seats per tenant — per-tenant licensing (S4)

ID	P	Test	Steps	Expected
TC-2F01	P1	Seat claimed against the employee's tenant key (Pre: Khan ACTIVE with free seats)	1. New Khan employee registers a device 2. Central licence page for Khan	Seat activates on Khan's licence at Central; devices_active increments for Khan only
TC-2F02	P1	Tenant seat limit blocks only that tenant (Pre: Tenant at device_limit)	1. One more device registers for that tenant 2. Khan (different tenant) registers a device	First gets 409 LICENSE_SEAT_BLOCKED device_limit_reached; Khan's registration succeeds
TC-2F03	P2	Unbind releases the tenant's seat (Pre: Bound device on Khan)	1. Devices → Unbind 2. Check Central seat count for Khan	Seat released on Khan's licence; token revoked; device blocked from re-registering until rebind

ID	P	Test	Steps	Expected
TC-2F04	P2	Rebind reclaims on the right licence (<i>Pre: Unbound Khan device</i>)	1. Devices → Rebind	Seat reclaimed against Khan's key; 409 with clear message if none free
TC-2F05	P2	Relieve employee releases every device per tenant (<i>Pre: Khan employee with 2 devices</i>)	1. Employees → Relieve (with reason)	Both seats released on Khan's licence; devices unbound; tokens revoked

2G. Storage reporting & governance — per-tenant licensing (S4)

ID	P	Test	Steps	Expected
TC-2G01	P1	Per-tenant storage reported to Central (<i>Pre: Two tenants with different storage footprints</i>)	1. Validate each tenant (Tenants → Validate) 2. Central: storage_usages rows	Each tenant's storage_gb equals ITS OWN files only (not the whole box)
TC-2G02	P2	Install row still reports whole-box storage (<i>Pre: Install slot keyed (client-hosted case)</i>)	1. Validate install licence	storage_gb = all evidence files on the server
TC-2G03	P1	Tenant over quota pauses only that tenant's screenshots (<i>Pre: Force tenant over 100% (Central storage governance)</i>)	1. Validate tenant (pause flag returns) 2. That tenant's agent uploads screenshot 3. Khan agent uploads screenshot	Over-quota tenant gets 409 'Storage is full...'; Khan uploads fine; activity tracking continues for both
TC-2G04	P2	Global storage pause still works (backward compat) (<i>Pre: Set Setting storage_paused=1</i>)	1. Any agent uploads screenshot	409 for all companies — install-wide switch honoured as before
TC-2G05	P2	Storage bar colours & percentages on Tenants screen (<i>Pre: Tenants at <70%, 70–89%, ≥90% of quota</i>)	1. Open Tenants; compare used/quota/%	Bar green <70, amber 70–89, red ≥90; % = used/quota rounded; 'no quota' shown when unlimited

2H. Tenants screen (HOST) — per-tenant licensing (S4)

ID	P	Test	Steps	Expected
TC-2H01	P1	Visible to Super Admin only (<i>Pre: Accounts of each role</i>)	1. Sign in as SUPER_ADMIN, COMPANY_ADMIN, HR_ADMIN, EMPLOYEE in turn	HOST → Tenants appears only for SUPER_ADMIN; other roles never see the section
TC-2H02	P1	API gated server-side (<i>Pre: Company admin token</i>)	1. GET /api/tenants 2. POST /api/tenants/{id}/license/validate	Both 403 regardless of UI

ID	P	Test	Steps	Expected
TC-2H03	P2	KPI chips correct (Pre: Known tenant mix)	1. Open Tenants; cross-check chips	Cloud tenants count, other companies, Licences OK x/y, devices bound, total evidence storage all match DB
TC-2H04	P2	Slug links open branded consoles (Pre: Tenants with slugs)	1. Click /khan link	New tab: admin.smartept.com/khan branded login/console
TC-2H05	P1	Licence pills reflect every state (Pre: Tenants in states: active, evaluation, no licence, expired; plus a non-cloud company)	1. Review each row's pill	ACTIVE green; EVALUATION amber with days left; NO LICENCE red; EXPIRED per grace; non-cloud shows INSTALL pill
TC-2H06	P2	Seats column warns when full (Pre: Tenant with bound == device_limit)	1. Review seats cell	Rendered amber/bold 'n / n'
TC-2H07	P1	Per-tenant Validate button works (Pre: Tenant with stale last check)	1. Click Validate on the row	Phone-home fires for THAT tenant only; button shows result; row refreshes with new last-check time
TC-2H08	P3	Storage cache notice	1. Footer note on Tenants	States 10-minute refresh + billing-lives-in-Central reminder + install licence status
TC-2H09	P2	Validate on non-cloud company rejected cleanly	1. POST /api/tenants/{nonCloudId}/license/validate	422 'uses the install-level licence — validate it from the Licence screen'

2I. Daily validate command — per-tenant licensing (S4)

ID	P	Test	Steps	Expected
TC-2I01	P1	Command validates install + every tenant row (Pre: Install + 2 tenant rows keyed)	1. php artisan smartept:validate-license	One output line per configured row (Install / Tenant #id (name)); each bundle refreshed
TC-2I02	P2	Exit code fails when any row blocked (Pre: One tenant revoked)	1. Run command; echo \$?	Non-zero exit; blocked row named in output
TC-2I03	P3	No keys anywhere → graceful warn (Pre: Fresh DB, no keys)	1. Run command	Warning about no key; exit success; nothing crashes

2J. Automated & regression — per-tenant licensing (S4)

ID	P	Test	Steps	Expected
TC-2J01	P1	PerTenantLicenseTest green (Pre: Local Laragon)	1. php artisan test --filter=PerTenantLicenseTest	All 5 tests pass
TC-2J02	P1	M9 licence + M11 offboarding suites green (regression)	1. php artisan test --filter=M9LicenseTest 2. --filter=M11OffboardingTest	All pass unchanged — client-hosted behaviour untouched
TC-2J03	P2	Full product suite green	1. test.bat	No failures
TC-2J04	P1	Live smoke after deploy (Pre: Steps 3–4 of SR-04 §12 done)	1. /khan web login 2. Khan agent sign-in + sync 3. Tenants screen on live 4. Central tenant storage stats next day	All green; Khan on own licence; per-tenant storage flowing

3A. Migration & automated tests — licence change flows (S5)

ID	P	Test	Steps	Expected
TC-3A01	P1	Migration adds renewal_device_limit cleanly (Pre: Central at face8e7; local Laragon DB)	1. php artisan migrate 2. DESCRIBE licences	2026_08_12_000500 runs clean; renewal_device_limit nullable UNSIGNED INT after device_limit; no ENUMs; no data lost
TC-3A02	P1	LicenceChangeFlowsTest green	1. php artisan test --filter=LicenceChangeFlowsTest	All 5 tests pass (incl. the exact ₹10,000 + ₹1,800 GST interpolated-difference case)
TC-3A03	P1	Progressive pricing regression green	1. php artisan test --filter=PerpetualProgressivePricingTest	Full acceptance matrix still passes — the upgrade engine changed nothing in base pricing
TC-3A04	P2	Migration rollback clean (Pre: Migration applied)	1. php artisan migrate:rollback --step=1 2. Re-migrate	Column drops and re-adds without error

3B. Admin billed upgrade — cloud subscription — licence change flows (S5)

ID	P	Test	Steps	Expected
TC-3B01	P1	Upgrade... button appears on active licences only (Pre: Licences list with active + suspended + trial rows)	1. Central admin → Licences 2. Compare rows	Active subscription & perpetual rows show Upgrade...; suspended/expired/trial rows do not

ID	P	Test	Steps	Expected
TC-3B02	P1	Pro-rata order raised with GST paper trail (<i>Pre: Active annual cloud subscription, e.g. 25 users, months remaining</i>)	1. Upgrade... → new total 30 → Raise upgrade order	Order number returned; description 'Mid-period upgrade 25 → 30 users — pro-rata for N remaining day(s)'; total = diff/month × remaining months + 18% GST; pay link copyable
TC-3B03	P1	Recording payment applies the new limit instantly (<i>Pre: Upgrade order from previous case</i>)	1. Billing → the order → Record payment (full) 2. Licences list	Licence shows the new limit; expiry date UNCHANGED; licence History shows upgrade_order + order paid; GST invoice issued in the shared series
TC-3B04	P2	Reduction attempt rejected with guidance (<i>Pre: Active 30-user subscription</i>)	1. Upgrade... → enter 20	422: 'Enter a user count higher than the current 30 — reductions apply at renewal, not mid-period'
TC-3B05	P2	Expired-period subscription refused (renew instead) (<i>Pre: Subscription past expiry</i>)	1. Upgrade... → any higher count	422: 'no remaining period — use Renew instead'

3C. Admin billed upgrade — perpetual — licence change flows (S5)

ID	P	Test	Steps	Expected
TC-3C01	P1	Exact interpolated difference on example bands (<i>Pre: Perpetual licence 15 users; bands 15 → ₹15k, 30 → ₹25k configured</i>)	1. Upgrade... on the perpetual row → 30 users → Raise order	Subtotal exactly ₹10,000; GST ₹1,800; total ₹11,800; description 'Lifetime licence upgrade 15 → 30 users — one-time price difference'
TC-3C02	P1	Payment raises limit; AMC re-bases next cycle (<i>Pre: Order from previous case</i>)	1. Record payment 2. Later: raise AMC quote for this client	device_limit = 30 immediately; the NEXT AMC quote prices 18% of the 30-user interpolated price (amcQuote reads prevailing count)
TC-3C03	P1	Above priced bands → custom quote, never a price (<i>Pre: Bands end at 100 users</i>)	1. Upgrade... → 150 users	422 custom-quotation message with WhatsApp number; NO order created; ₹0 impossible
TC-3C04	P2	Perpetual reduction rejected (<i>Pre: Perpetual 30 users</i>)	1. Upgrade... → 20	422: 'a lifetime licence never reduces once purchased'
TC-3C05	P2	Modal wording is kind-aware	1. Open Upgrade... on a subscription, then on a perpetual	Subscription modal explains pro-rata/expiry-unchanged; perpetual modal explains one-time difference + .lic re-issue note

3D. Downgrade-at-renewal — admin side — licence change flows (S5)

ID	P	Test	Steps	Expected
TC-3D01	P1	Edit licence schedules a reduction (Pre: Active 40-user subscription)	1. Edit → 'Reduce to (at next renewal)' = 25 → Save 2. Licences list	Saved; row shows amber '↓ 25 at renewal'; History shows licence.edited with renewal_device_limit
TC-3D02	P1	Guard: not below devices in use (Pre: Licence with 10 active devices)	1. Edit → reduction = 5 → Save	422: 'Scheduled reduction is below the devices currently in use — free seats first'
TC-3D03	P2	Blank clears the schedule (Pre: Schedule set)	1. Edit → clear the field → Save	renewal_device_limit NULL; badge gone

3E. Downgrade-at-renewal — portal + billing — licence change flows (S5)

ID	P	Test	Steps	Expected
TC-3E01	P1	Portal card schedules a reduction (Pre: Client portal login; active 40-user subscription; 3 devices in use)	1. Buy & Renew → 'Reduce users at next renewal' → 25 → Schedule	Toast confirms; card shows 'Scheduled: your next renewal reduces to 25 users'; Cancel button appears; audit client.schedule_reduction
TC-3E02	P1	Renewal order bills the REDUCED size (Pre: Schedule = 25 on a 40-user licence)	1. Portal → Renew now (or admin raises renewal) 2. Inspect order	Order line priced for 25 devices; description contains '25 devices' + 'scheduled reduction from 40 users applies on payment'; meta apply_device_limit = 25
TC-3E03	P1	Payment applies reduction and clears schedule (Pre: Renewal order from previous case)	1. Record payment / pay 2. Check licence	device_limit = 25; renewal_device_limit = NULL; expiry extended one period; portal card no longer shows a schedule
TC-3E04	P2	Portal guard: not a reduction / below devices in use (Pre: 40-user licence, 3 devices in use)	1. Try scheduling 45 2. Try scheduling 2	45 → 'That is not a reduction — use the mid-period upgrade'; 2 → 'You currently have 3 device(s) in use...'
TC-3E05	P2	Cancel schedule works (Pre: Schedule set)	1. Portal card → Cancel schedule	renewal_device_limit NULL; message 'renewals bill your current size again'
TC-3E06	P2	Perpetual cannot schedule a reduction (Pre: Perpetual licence id)	1. POST client/api/licences/{id}/schedule-reduction devices=10	422: 'Only Cloud subscriptions reduce at renewal — a lifetime licence never reduces once purchased'

ID	P	Test	Steps	Expected
TC-3E07	P3	Renewal without schedule unchanged (regression) (Pre: No schedule)	1. Renew and pay	Billed at current size; limit unchanged; identical to pre-change behaviour

3F. Portal upgrade cards & visibility — licence change flows (S5)

ID	P	Test	Steps	Expected
TC-3F01	P1	Perpetual tenant sees lifetime upgrade card (Pre: Portal login of a perpetual client (active))	1. Buy & Renew	'Add users to your lifetime licence' card with one-time-difference wording + never-reduces note; Get upgrade amount & pay works end-to-end via pay page
TC-3F02	P1	Cloud tenant still sees pro-rata card (regression) (Pre: Active cloud subscription with days remaining)	1. Buy & Renew	'Add users mid-period (pro-rata)' card exactly as before; payment raises seats instantly
TC-3F03	P2	Why-not note replaces silence (Pre: Trial tenant portal; then an expired tenant)	1. Buy & Renew on each	Trial: 'Trials upgrade by buying a plan below...'; expired: 'Your licence is expired — renew or contact us...' — never a blank page
TC-3F04	P2	Portal upgrade endpoint branches by kind (Pre: Perpetual licence via API)	1. POST client/api/licences/{id}/upgrade devices=30	Order created via createPerpetualUpgradeOrder (description says lifetime price difference), not the pro-rata engine

3G. Self-service .lic re-download — licence change flows (S5)

ID	P	Test	Steps	Expected
TC-3G01	P1	Bound licence downloads a fresh signed .lic (Pre: Client-hosted active licence WITH server_fingerprint; portal login)	1. Portal → Licences → Download licence file (.lic) 2. Import on the product Licence screen	File downloads (correct filename); product imports and activates; bundle carries CURRENT device_limit; audit licence.file_issued by client:
TC-3G02	P1	Unbound licence gets guidance, not a file (Pre: Client-hosted licence with NO fingerprint)	1. Portal → Licences	Message: 'available once your server has validated online once (or send us its fingerprint...)'; endpoint returns 422 if called directly

ID	P	Test	Steps	Expected
TC-3G03	P1	Client can NEVER set a new fingerprint (<i>Pre: Bound licence</i>)	1. POST client/api/licences/{id}/license-file with a 'fingerprint' body field	Field ignored — file is signed with the STORED fingerprint only; moving machines still requires admin Shift machine
TC-3G04	P2	Cloud-deployment client refused politely (<i>Pre: Cloud tenant portal</i>)	1. Call the license-file endpoint	422: 'Licence files are for self-hosted servers — your Cloud console is licensed automatically'
TC-3G05	P2	Post-upgrade .lic carries the new seat count (<i>Pre: Perpetual client upgraded 15 → 30 and paid</i>)	1. Re-download .lic 2. Import on server	Product Licence screen shows 25 → 30-seat entitlement (new device_limit) after import
TC-3G06	P2	Cross-tenant access blocked (<i>Pre: Two tenants; tenant B's licence id</i>)	1. As tenant A, call license-file / upgrade / schedule-reduction on B's id	404 on every endpoint (tenant scoping)
TC-3G07	P3	Admin Licence file button unchanged (regression)	1. Admin → Licences → Licence file (with and without stored fingerprint)	Behaviour identical to 6-Aug build incl. fingerprint remembering and re-download

3H. Paper trail & money regression — licence change flows (\$5)

ID	P	Test	Steps	Expected
TC-3H01	P1	GST invoice issued in the shared series for upgrades (<i>Pre: Paid admin upgrade order</i>)	1. Billing → Invoices	Invoice numbered in the single EPT-{FY} series; amounts match order (subtotal/GST/total); appears in client portal invoices
TC-3H02	P1	History timeline complete (<i>Pre: Licence that was upgraded + scheduled + .lic-downloaded</i>)	1. Admin → Licences → History	Timeline shows upgrade_order, order paid, licence.edited (schedule), file_issued (client) — nothing missing
TC-3H03	P2	USD tenant upgrade converts currency (<i>Pre: Tenant with currency USD</i>)	1. Raise an upgrade order	Line converted at usd_inr_rate; GST 0 for USD; '(USD)' suffix on the line
TC-3H04	P2	Existing flows untouched (Central smoke)	1. php artisan test (full Central suite) or test.bat	FinancialFlowTest, GstInvoiceTest, ClientPortalTest, CentralSmokeTest all green

Automated companions: product *PerTenantLicenseTest* + full M-suites; Central *LicenceChangeFlowsTest* + *PerpetualProgressivePricingTest* + full suite (56 tests / 342 assertions, green).